School of Natural Sciences and Mathematics

Department of Sustainable Earth Systems Sciences

Geosciences (BS)

Attaining a fundamental understanding of Earth's natural systems, past and present, and their coupled nonlinear relationships, is a primary goal of geosciences. An earth scientist studies the Earth's structure, composition and evolution, its minerals, rocks, fossils, sediments and fluids, and investigates the geological, physical, and biogeochemical processes occurring within the Earth and on its surface.

Professional career opportunities in earth sciences are available in the environmental, energy, mineral, and groundwater resource industries, and in government, educational, and research organizations related to these fields. In addition, many occupations concerned with business, management, economics, law, policy, and engineering, require a background in earth sciences.

Specific degree plans will be formulated by the undergraduate advisor in the Department of Sustainable Earth Systems Sciences. Changing circumstances may require changes to the degree plans.

Bachelor of Science in Geosciences

Degree Requirements (120 semester credit hours)

View an Example of Degree Requirements by Semester

Faculty

FACG> nsm-geosciences-bs

Professors: David Hyndman, David Lumley, Robert J. Stern

Assistant Professors: Kristina Butler, Nadine Igonin, Zachary Sickmann, Xian Wu, Qian Zhao, Yujie Zheng

Professors Emeriti: John F. Ferguson, John W. Geissman, William I. Manton, George A. McMechan

Professors of Instruction: Ignacio Pujana

Associate Professor of Instruction: William R. Griffin, Steven Pirouz

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

Select any 6 semester credit hours from Communication Core courses (see advisor)

Mathematics: 3 semester credit hours

MATH 2413 Differential Calculus[1, 2]

or MATH 2417 Calculus I[1, 2]

Or select any 3 semester credit hours from Mathematics Core courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

CHEM 1311 General Chemistry I[1]

or CHEM 1315 Honors Freshman Chemistry I[1]

CHEM 1312 General Chemistry II[1]

or CHEM 1316 Honors Freshman Chemistry II[1]

Or select any 6 semester credit hours from Life and Physical Sciences Core courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from Language, Philosophy and Culture Core courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from Creative Arts Core courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from American History Core courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from Government/Political Science Core courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from Social and Behavioral Sciences Core courses

Component Area Option: 6 semester credit hours

GEOS 1303 Physical Geology

or GEOS 2310 Environmental Geosciences

GEOS 1304 History of Earth and Life

Or select any 6 semester credit hours from Component Area Option Core courses (see advisor)

II. Major Requirements: 63-78 semester credit hours

Major Preparatory Courses: 22-26 semester credit hours beyond Core Curriculum

NATS 1101 Natural Sciences and Mathematics Freshman Seminar

Prerequisite courses to be completed before enrolling in upper-division GEOS courses.

CHEM 1111 General Chemistry Laboratory I

or CHEM 1115 Honors Freshman Chemistry Laboratory I

CHEM 1112 General Chemistry Laboratory II

or CHEM 1116 Honors Freshman Chemistry Laboratory II

CHEM 1311 General Chemistry I[1]

or CHEM 1315 Honors Freshman Chemistry I[1]

CHEM 1312 General Chemistry II[1]

or CHEM 1316 Honors Freshman Chemistry II[1]

GEOS 1303 Physical Geology[1]

or GEOS 1305 Our Blue Planet

or GEOS 1306 Geohazards

or GEOS 2310 Environmental Geosciences[1]

GEOS 1103 Earth Science Laboratory

GEOS 1304 History of Earth and Life[1]

GEOS 1104 History of Earth and Life Laboratory

GEOS 2411 Earth Materials

MATH 2413 Differential Calculus[1,2]

or MATH 2417 Calculus I[1,2]

MATH 2414 Integral Calculus

or MATH 2419 Calculus II

PHYS 2325 Mechanics

and PHYS 2125 Physics Laboratory I

or PHYS 2421 Honors Physics I - Mechanics and Heat[3]

PHYS 2326 Electromagnetism and Waves

or PHYS 2422 Honors Physics II - Electromagnetism and Waves

PHYS 2126 Physics Laboratory II

Major Core Courses: 27 semester credit hours

GEOS 3421 Sedimentary Systems

GEOS 3470 Structural Geology

GEOS 3475 Geochemistry

GEOS 4300 Earth Science Field School

GEOS 4301 Geology Field School

GEOS 4320 The Physics of the Solid Earth

GEOS 4390 Communication in Earth Sciences

And select one course from the following:

GEOS 4310 Environmental Geophysics

GEOS 4395 Remote Sensing Geophysics

Students may select either the Geology Option or the Geophysics Option.

A. Geology Option: 14-15 semester credit hours

GEOS 3434 Paleobiology

GEOS 4322 The Earth System

GEOS 4430 Hydrogeology and Aqueous Geochemistry

And select one mathematics course from the following:

GEOS 3351 Mathematics in the Earth Sciences

ENGR 2300 Linear Algebra for Engineers

MATH 2415 Calculus of Several Variables

MATH 2418 Linear Algebra

MATH 3351 Advanced Calculus

PHYS 3330 Numerical Methods in Physics and Computational Techniques

STAT 3332 Statistics for Life Sciences

STAT 3355 Introduction to Data Analysis

B. Geophysics Option: 25 semester credit hours

MATH 2418 Linear Algebra

MATH 2420 Differential Equations with Applications

MATH 3351 Advanced Calculus

MATH 4362 Partial Differential Equations

PHYS 3411 Theoretical Physics

PHYS 3312 Classical Mechanics

PHYS 3416 Electricity and Magnetism

III. Elective Requirements: 0-15 semester credit hours

Both lower- and upper-division courses may count as electives.

- 11-15 semester credit hours for Geology Option
- 0-4 semester credit hours for Geophysics Option

Students are strongly encouraged to consider taking GEOS graduate courses as free electives if they meet the fast-track requirements.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. A Major requirement that also fulfills a Core Curriculum requirement.

2. Three semester credit hours are counted to fulfill the Mathematics Core Requirement with the remaining semester credit hour to be counted under the major requirements.

3. Students who complete PHYS 2421 do not need to complete PHYS 2125.

UTeach Option

The UTeach option may be added to either the BA or BS degree. UTeach Dallas Option degree plans are streamlined to allow students to complete both a rigorous Bachelor of Science or Bachelor of Arts degree and all coursework for middle or high school teacher certification in four years. Teaching Option degrees require deep content knowledge combined with courses grounded in the latest research on math and science education. While most graduates go on to classroom teaching, UTeach alums are also prepared to enter graduate school and to work in a discipline related industry.

Fast Track Baccalaureate/Master's Degrees

An accelerated BS/MS Fast Track Program is available for undergraduate students with strong academic records who intend to pursue master's studies at UT Dallas. After Fast Track admission to an MS program, students may take up to 15 semester credit hours of approved graduate courses in their senior year to use toward completion of both the BS and MS degrees. Students must have completed 90 or more semester credit hours (out of which 36 hours are from the core curriculum), with a cumulative GPA of at least 3.00. In addition, students must have completed CHEM 1311 and CHEM 1312 with labs, MATH 2413 and MATH 2414, and PHYS 2325 and PHYS 2326 with labs; or equivalent courses.

Interested students, after reviewing the UT Dallas Fast Track policy, should contact their undergraduate advisor and the graduate advisor of the intended MS program.

Minors

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

For all minors in the School of Natural Sciences and Mathematics students must complete all prerequisite sequences for required minor courses.

Minor in Geosciences

20 semester credit hours

Required lower-division courses: 8 semester credit hours

GEOS 1303 Physical Geology[4]

or GEOS 1305 Our Blue Planet

or GEOS 1306 Geohazards

or GEOS 2310 Environmental Geosciences

GEOS 1103 Earth Science Laboratory[4]

GEOS 1304 History of Earth and Life[4]

GEOS 1104 History of Earth and Life Laboratory[4]

Upper-division courses: 12 semester credit hours

To be selected in consultation with department Faculty Undergraduate Advisor

4. A prerequisite course to be completed before enrolling in upper-division GEOS core courses (GEOS 4301, GEOS 3421, GEOS 3434, GEOS 3464, GEOS 3470, GEOS 4300, GEOS 4320, GEOS 4322, and GEOS 4430).

Certificates

Certificate in Geophysics

12 semester credit hours

A Certificate in Geophysics is offered by the Department of Sustainable Earth Systems Sciences in the School of Natural Sciences and Mathematics.

The focus of the Certificate in Geophysics is to provide students with the opportunity to combine their existing knowledge with a strong foundation and understanding in Geophysics, which will open up exciting new job opportunities for their future careers in industry, government, and academia.

Admission Requirements

Undergraduate students who have completed prerequisite courses.

Certificate Requirements

Students must complete the following courses:

GEOS 4310 Environmental Geophysics

GEOS 4320 The Physics of the Solid Earth

GEOS 4395 Remote Sensing Geophysics

or GISC 4395 Remote Sensing Geophysics

Choose one course from the following:

GEOS 2307 Digital Earth

ISNS 2366 Global Climate Change

PHYS 3380 Astronomy

Certificate in Sustainable Earth and Environment

12-13 semester credit hours

A Certificate in Sustainable Earth and Environment is offered by the Department of Sustainable Earth Systems Sciences in the School of Natural Sciences and Mathematics.

The focus of the Certificate in Sustainable Earth and Environment is to provide students with a basic foundational understanding and technical skills in earth and environmental science, combined with "Sustainable Earth" aspects of natural resources, economics, business, social science, and health. The certificate is designed with a two-course core giving a basic overview of earth and environmental science, and a second set of courses that provides technical skills in quantitatively measuring current environments. An expansive third set of electives provides further breadth in areas of special interest for each student.

Admission Requirements

This Certificate is open to all UT Dallas students enrolled in an undergraduate degree program.

Geosciences students cannot use more than 2 degree-required courses to fulfill both certificate and B.S. requirements.

Certificate Requirements

Core Courses: 6 semester credit hours

GEOS 1303 Physical Geology

GEOS 2310 Environmental Geosciences

Technical Application Courses: 3 semester credit hours

Choose one course from the following:

GEOS 2309 Principles of Geospatial Information Sciences

or GISC 2309 Principles of Geospatial Information Sciences

GEOS 4310 Environmental Geophysics

GEOS 4325 Introduction to Remote Sensing

or GISC 4325 Introduction to Remote Sensing

GEOS 4395 Remote Sensing Geophysics

or GISC 4395 Remote Sensing Geophysics

Breadth Courses: 3-4 semester credit hours

Choose one course from the following:

GEOS 2302 The Global Environment

or GEOG 2302 The Global Environment

GEOS 2304 Earth Energy Resources

GEOS 2321 Geology, Resources, and Environment of Latin America

GEOS 3421 Sedimentary Systems

GEOS 4430 Hydrogeology and Aqueous Geochemistry

GEOS 2307 Digital Earth

or GISC 2307 Digital Earth

GISC 3336 Fundamentals of Sustainability

or ENVR 3336 Fundamentals of Sustainability

or EPPS 3336 Fundamentals of Sustainability

GISC 4386 Climate Change and Sustainable Solutions

or ENVR 4386 Climate Change and Sustainable Solutions

or EPPS 4386 Climate Change and Sustainable Solutions

ISNS 2366 Global Climate Change

ENGY 3302 Managing Power and Renewable Energy Assets; Sustainability

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